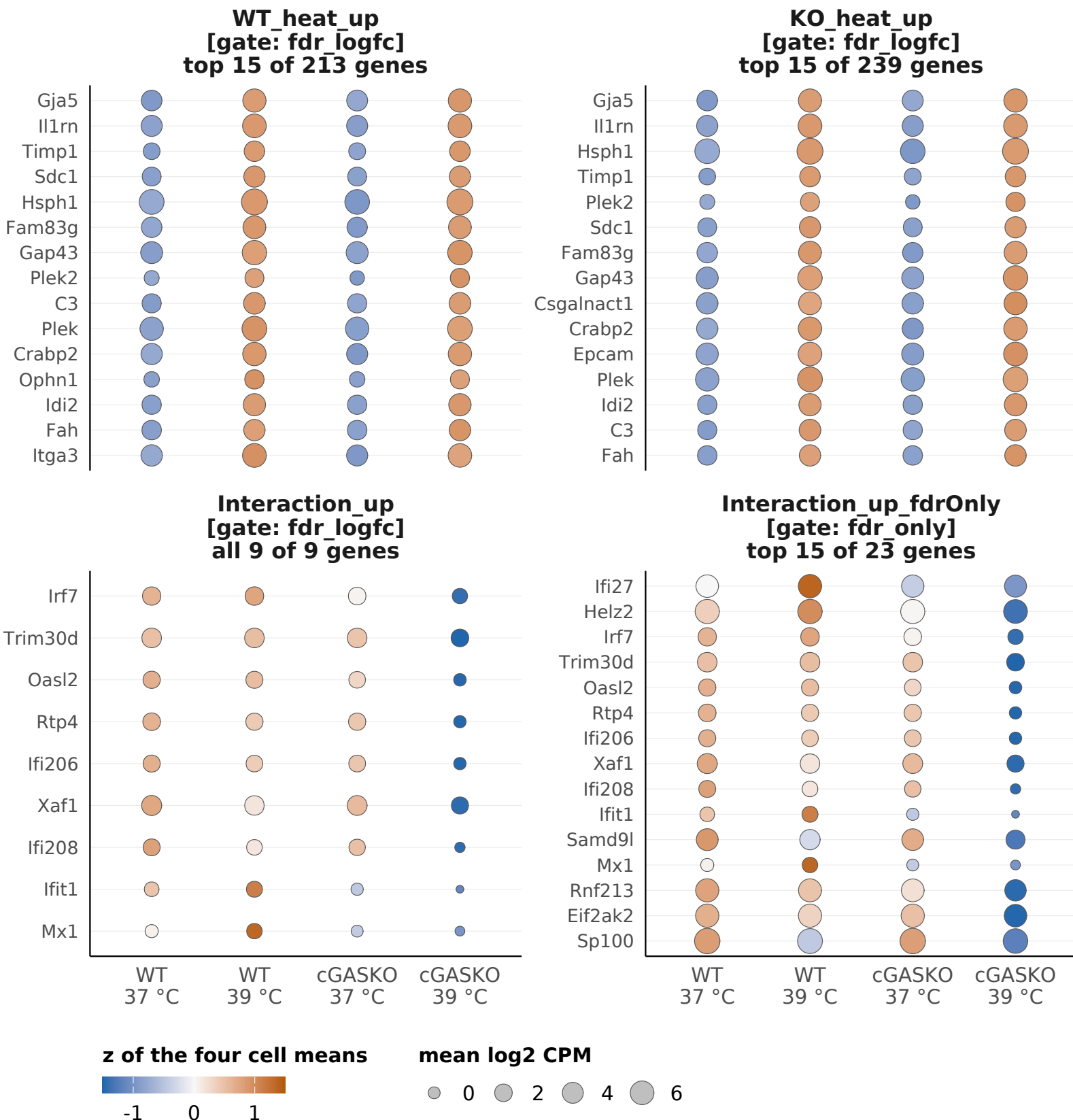


Signature genes across the 2x2 design (up arms)

Each exported UP arm read back gene by gene in the four design cells, ordered by the arm's own source-contrast t. Down arms are outside this stage's scope.



Group mean of log2 CPM over n=5 libraries per design cell (01_eda). Fill = the gene's four cell means z-scored across those cells (bounded at ± 1.5 for four cells); size = mean log2 CPM. Gate: adj.P.Val < 0.05 (fdr_only), or additionally $|\log_2\text{FC}| \geq 1$ (fdr_logfc). Panels show each arm's top 15 genes by t, or all of them where an arm has fewer — the strip says which, and the stage tables carry every member. Mouse symbols. Claim tier: L3 (n=5/group). Sample mapping owner-confirmed (GSE329522 iTreg 2x2 genotype x temperature).